

Branch-Indexed Multiplication and Logarithm Algebra on the Universal Cover of $\mathbb{C}^\times$

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Abstract

We formalize the branch-index layer used in the Complex Numeric Representational System (CNRS) as a concrete model of the universal covering group of the nonzero complex numbers. A lifted value is represented by a pair (z, k) with $z \in \mathbb{C}^\times$ and $k \in \mathbb{Z}$, where the lifted argument is $\text{Arg } z + 2\pi k$. We derive the exact branch-wrap cocycle governing multiplication, prove associativity, commutativity, inversion, and integer-power laws, and show that the lifted logarithm is a single-valued group isomorphism from the covering group to the additive complex plane. We also distinguish this multiplicative algebra from addition: ordinary complex addition does not induce a canonical globally defined group law on the cover of $\mathbb{C}^\times$. The resulting theorem package gives a precise mathematical foundation for CNRS branch-index multiplication and logarithmic bookkeeping while remaining independent of CNRS-A digit representation.

1 Purpose and status

CNRS Layer 2 records the sheet of the complex logarithm by adjoining an integer branch index to a nonzero complex value. The purpose of this note is to replace informal branch bookkeeping by a complete algebraic construction.

The results are standard consequences of covering-space theory, but the explicit pair model, cocycle, and computational formulas are the structures required by CNRS. The contribution here is therefore an exact alignment theorem between the CNRS branch-index representation and the universal covering group of $\mathbb{C}^\times$.

Throughout, the principal argument is

$$\text{Arg } z \in (-\pi, \pi],$$

and the principal logarithm is

$$\text{Log } z = \ln |z| + i \text{Arg } z.$$

The point $z = 0$ is excluded.

2 Two equivalent models of the cover

Definition 2.1 (Angle model). *Define*

$$\mathcal{U} = \mathbb{R} \times \mathbb{R}$$

with group law

$$(u, \theta) + (v, \phi) = (u + v, \theta + \phi)$$

and covering map

$$p(u, \theta) = e^{u+i\theta} \in \mathbb{C}^\times.$$

This is the standard simply connected universal cover of $\mathbb{C}^\times$.

Definition 2.2 (Branch-index model). *Define*

$$\widetilde{\mathbb{C}^\times} = \mathbb{C}^\times \times \mathbb{Z}.$$

For $(z, k) \in \widetilde{\mathbb{C}^\times}$, its lifted argument and lifted logarithm are

$$\Theta(z, k) = \text{Arg } z + 2\pi k, \quad \widetilde{\text{Log}}(z, k) = \ln |z| + i\Theta(z, k).$$

Proposition 2.3. *The map*

$$F : \widetilde{\mathbb{C}^\times} \longrightarrow \mathcal{U}, \quad F(z, k) = (\ln |z|, \text{Arg } z + 2\pi k)$$

is a bijection.

Proof. Injectivity follows because equality of the first coordinates gives equal moduli, while equality of lifted angles gives equal arguments modulo 2π and hence equal complex values; the principal-argument convention then forces equal branch indices. For surjectivity, given (u, θ) let $z = e^{u+i\theta}$ and let

$$k = \frac{\theta - \text{Arg } z}{2\pi} \in \mathbb{Z}.$$

Then $F(z, k) = (u, \theta)$. □

3 The branch-wrap cocycle

Definition 3.1 (Wrap cocycle). *For $z, w \in \mathbb{C}^\times$ define*

$$\omega(z, w) = \frac{\text{Arg } z + \text{Arg } w - \text{Arg}(zw)}{2\pi}.$$

Lemma 3.2. *For all $z, w \in \mathbb{C}^\times$, $\omega(z, w) \in \mathbb{Z}$. With the convention $\text{Arg} \in (-\pi, \pi]$, one has*

$$\omega(z, w) \in \{-1, 0, 1\}.$$

Proof. The two quantities $\text{Arg } z + \text{Arg } w$ and $\text{Arg}(zw)$ represent the same angle modulo 2π , so their difference is an integer multiple of 2π . Since the first lies in $(-2\pi, 2\pi]$ and the second in $(-\pi, \pi]$, only the three displayed integers can occur. □

An explicit rule is

$$\omega(z, w) = \begin{cases} 1, & \text{Arg } z + \text{Arg } w > \pi, \\ -1, & \text{Arg } z + \text{Arg } w \leq -\pi, \\ 0, & \text{otherwise,} \end{cases}$$

where the lower boundary reflects the half-open principal interval.

Theorem 3.3 (Cocycle identity). *For all $z, w, u \in \mathbb{C}^\times$,*

$$\omega(z, w) + \omega(zw, u) = \omega(w, u) + \omega(z, wu).$$

Proof. Both sides equal

$$\frac{\text{Arg } z + \text{Arg } w + \text{Arg } u - \text{Arg}(z w u)}{2\pi}$$

after cancellation of the intermediate principal arguments. □

4 Lifted multiplication

Definition 4.1 (Branch-index multiplication). For $(z, k), (w, \ell) \in \widetilde{\mathbb{C}^\times}$, define

$$(z, k) \odot (w, \ell) = (zw, k + \ell + \omega(z, w)).$$

Theorem 4.2 (Covering-group algebra). The set $(\widetilde{\mathbb{C}^\times}, \odot)$ is an abelian group. Its identity is $(1, 0)$, and the projection

$$\pi : \widetilde{\mathbb{C}^\times} \rightarrow \mathbb{C}^\times, \quad \pi(z, k) = z$$

is a surjective group homomorphism with kernel

$$\ker \pi = \{(1, k) : k \in \mathbb{Z}\} \cong \mathbb{Z}.$$

Proof. Closure is immediate. Associativity follows from the cocycle identity. Commutativity follows from $zw = wz$ and symmetry of ω . Since $\text{Arg } 1 = 0$, $(1, 0)$ is an identity. The projection is visibly a homomorphism and is surjective. Its kernel is exactly the displayed set. $\square$

Proposition 4.3 (Inverse). Let $x = (z, k)$. Then

$$x^{-1} = (z^{-1}, -k - \omega(z, z^{-1})).$$

Equivalently, if $\text{Arg } z \neq \pi$ then

$$x^{-1} = (z^{-1}, -k),$$

while for negative real z (so $\text{Arg } z = \pi$),

$$x^{-1} = (z^{-1}, -k - 1).$$

Proof. The formula is forced by requiring the second component of $x \odot x^{-1}$ to vanish. The boundary case occurs because $\text{Arg } z = \text{Arg } z^{-1} = \pi$, so $\omega(z, z^{-1}) = 1$. $\square$

Proposition 4.4 (Integer powers). For $n \in \mathbb{Z}$, the power $x^{\odot n}$ is uniquely defined by the group law and satisfies

$$\Theta(x^{\odot n}) = n\Theta(x), \quad \pi(x^{\odot n}) = z^n.$$

Consequently

$$x^{\odot n} = \left(z^n, \frac{n(\text{Arg } z + 2\pi k) - \text{Arg}(z^n)}{2\pi} \right).$$

5 The single-valued lifted logarithm

Theorem 5.1 (Logarithm isomorphism). The lifted logarithm

$$\widetilde{\text{Log}} : (\widetilde{\mathbb{C}^\times}, \odot) \longrightarrow (\mathbb{C}, +), \quad \widetilde{\text{Log}}(z, k) = \ln |z| + i(\text{Arg } z + 2\pi k),$$

is a group isomorphism.

Proof. For $x = (z, k)$ and $y = (w, \ell)$,

$$\begin{aligned} \widetilde{\text{Log}}(x \odot y) &= \ln |zw| + i(\text{Arg}(zw) + 2\pi(k + \ell + \omega(z, w))) \\ &= \ln |z| + \ln |w| + i(\text{Arg } z + \text{Arg } w + 2\pi k + 2\pi \ell) \\ &= \widetilde{\text{Log}}(x) + \widetilde{\text{Log}}(y). \end{aligned}$$

Bijectivity follows from the earlier bijection F and the identification $(u, \theta) \mapsto u + i\theta$. $\square$

Corollary 5.2 (Exact product law). *For all $x, y \in \widetilde{\mathbb{C}^\times}$,*

$$\widetilde{\text{Log}}(x \odot y) = \widetilde{\text{Log}}(x) + \widetilde{\text{Log}}(y)$$

without an unspecified $2\pi i n$ term.

Corollary 5.3 (Lifted exponential). *Define*

$$\widetilde{\text{Exp}}(a + ib) = \left(e^{a+ib}, \frac{b - \text{Arg}(e^{a+ib})}{2\pi} \right).$$

Then

$$\widetilde{\text{Log}} \circ \widetilde{\text{Exp}} = \text{id}_{\mathbb{C}}, \quad \widetilde{\text{Exp}} \circ \widetilde{\text{Log}} = \text{id}_{\widetilde{\mathbb{C}^\times}}.$$

Thus the familiar many-valued logarithm is replaced on the cover by a globally single-valued inverse of the exponential map.

6 Fractional powers

For $x \in \widetilde{\mathbb{C}^\times}$ and $a \in \mathbb{C}$, define the lifted power

$$x^{[a]} = \widetilde{\text{Exp}}(a \widetilde{\text{Log}} x).$$

This is single-valued on the cover. It obeys

$$x^{[a+b]} = x^{[a]} \odot x^{[b]},$$

and for fixed a ,

$$(x \odot y)^{[a]} = \widetilde{\text{Exp}}(a \widetilde{\text{Log}} x + a \widetilde{\text{Log}} y).$$

When $a \in \mathbb{Z}$, this agrees with group powers. Projecting to $\mathbb{C}^\times$ recovers one branch-selected value of the usual multivalued power.

7 Why addition is different

Multiplication preserves $\mathbb{C}^\times$ and is the group operation whose universal cover is being constructed. Addition has three obstructions.

First, $z + w$ may be zero, which lies outside $\mathbb{C}^\times$. Second, a branch index records winding around zero under multiplication; it carries no canonical information about a path from z and w to $z + w$. Third, any rule

$$(z, k) \boxplus (w, \ell) = (z + w, m)$$

requires a choice of m that is not determined by the endpoint $z + w$ and the two individual sheet indices alone in a way compatible with arbitrary continuous deformations through the punctured plane.

Proposition 7.1 (No global lifted additive group). *There is no everywhere-defined binary operation $\boxplus$ on $\widetilde{\mathbb{C}^\times}$ for which:*

1. $\pi(x \boxplus y) = \pi(x) + \pi(y)$ for all x, y ;

2. $(\widetilde{\mathbb{C}^\times}, \boxplus)$ is a group;

3. π maps into $\mathbb{C}^\times$.

Proof. Choose y with $\pi(y) = -\pi(x)$. Then condition 1 requires $\pi(x \boxplus y) = 0$, contradicting the codomain $\mathbb{C}^\times$. $\square$

A partial lifted addition can be defined on pairs with $z + w \neq 0$ after selecting a path or a branch convention, but it is not intrinsic to the universal covering group and should not be conflated with the multiplication law proved above.

8 CNRS interpretation

The branch-index layer can now be stated precisely:

- CNRS-A may represent the projected value z numerically.
- The integer k records the logarithmic sheet.
- Multiplication combines the projected values and updates k by the wrap cocycle.
- The lifted logarithm is single-valued and additive.
- Inversion, powers, and roots are exact operations on the cover.
- Addition remains ordinary complex addition at the projected level and requires separate branch/path handling if a lifted result is desired.

The construction does not depend on the base $-2 + i$ or on any unresolved CNRS completeness theorem. It is a general topological-algebraic layer that can be paired with any exact or approximate representation of nonzero complex values.

9 Implementation specification

A computational implementation should store

$$(z, k), \quad z \neq 0,$$

with a single locked principal-argument convention. It should expose:

```
wrap_index(z, w)
lifted_multiply((z, k), (w, l))
lifted_inverse((z, k))
lifted_power_integer((z, k), n)
lifted_log((z, k))
lifted_exp(value)
lifted_power((z, k), exponent)
```

The implementation must test the branch-cut boundary, especially negative real inputs, because $\text{Arg}(-r) = \pi$ rather than $-\pi$ under the convention used here.

10 Claim status and open questions

The covering-group, cocycle, inverse, power, and logarithm results are established mathematical results in the explicit CNRS pair model. They complete the multiplicative/logarithmic algebra of the branch-index layer.

The following remain separate questions:

1. integration of the pair model with canonical CNRS-A representations;
2. exact branch-aware addition under specified path data;
3. analytic continuation of CNRS-H functions on the cover;
4. extension to matrices, operators, or fields with zeros and singularities;
5. physical interpretation of branch indices in Scale Space applications.

Conclusion

The CNRS branch index is not merely an auxiliary label. With the wrap cocycle it gives an exact coordinate model of the universal covering group of $\mathbb{C}^\times$. On this cover, multiplication is globally well defined, the logarithm is single-valued and additive, and inversion and powers have exact branch-aware formulas. This closes the core Layer-2 multiplication/logarithm problem while showing precisely why addition belongs to a different structure.

References

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AI assistance declaration

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